Stability of Multiagent Systems With Time-Dependent Communication Links

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***Abstract*—We study a simple but compelling model of network of agents interacting via time-dependent communication links. The model finds application in a variety of fields including synchronization, swarming and distributed decision making. In the model, each agent updates his current state based upon the current information received from neighboring agents. Necessary and/or sufficient conditions for the convergence of the individual agents' states to a common value are presented, thereby extending recent results reported in the literature. The stability analysis is based upon a blend of graph-theoretic and system-theoretic tools with the notion of convexity playing a central role. The analysis is integrated within a formal framework of set-valued Lyapunov theory, which may be of independent interest. Among others, it is observed that more communication does not necessarily lead to faster convergence and may eventually even lead to a loss of convergence, even for the simple models discussed in the present paper.**

***Index Terms*—Multiagent systems, set-valued Lyapunov theory, stability analysis, swarms, synchronization.**

I. Introduction

RECENT years have witnessed an increasing interest in the interaction between information flow and system dynamics. It is further recognized that information and communication constraints may have a considerable impact on the performance of a control system. The study of these topics forms an active area of research, giving rise to new control paradigms such as *quantized control systems*, *networked control systems* and *multiagent (multivehicle, multirobot) systems*.

An important aspect of information flow in a dynamical system is the communication topology which determines what information is available for which component at a given time instant. We study the role of the communication topology in the context of multi-agent systems. Our interest in multi-agent systems is motivated by the emergence of several applications including formation flying of unmanned aerial vehicles (UAVs), cooperative robotics [2]–[4] and sensor networks [5], [6].

In this paper, we consider a group of n agents, not necessarily identical. The individual agents share a common state space and each agent updates his current state based upon the information received from other agents, according to a simple rule. Our aim is to relate the information flow and communication structure with the stability properties of the group of agents.

The model that we consider encompasses, or is closely related to, several models reported in the literature. A prominent and well-studied example concerns synchronization of coupled oscillators, a phenomenon which is ubiquitous in the natural world and finds several applications in physics and engineering; see [7] for a review and a historical perspective. The Kuramoto equation [8]–[10], which is widely appreciated in that field, models a population of oscillators with (typically sinusoidal) coupling terms. The Kuramoto equation may be studied with the tools of this paper, at least when attention is restricted to identical oscillators whose initial phase differences are sufficiently small. Another interesting example concerns swarming, a cooperative behavior observed for a variety of living beings such as birds, fish, bacteria, etc. In the physics literature, swarming models are often individual-based with each individual being represented by a particle moving at constant speed, its direction of motion being updated according to nearest neighbor coupling; see, for example, [11]. In recent years, engineering applications such as formation control have increased the interest of engineers in swarming and collective motion patterns [12]–[26]. The approach of the present paper applies to the linear swarming model of [14] as well as to the nonlinear model of [11] provided the initial orientation differences are sufficiently small. A third and last example of multiagent systems that fall within the scope of this paper are the consensus algorithms studied in [27]–[31]. Consensus protocols enable a network of agents to agree upon quantities of interest via a process of distributed decision making. Distributed agreement problems have a long history; see, for example, the book [32, Sec. 4.6] and [33].

We impose very weak assumptions on the communication topology; we allow for nonbidirectional and time-dependent communication patterns. Unidirectional communication is important in practical applications and can easily be incorporated, for example, via broadcasting. Also, sensed information flow which plays a central role in schooling and flocking is typically not bidirectional. In addition, we do not exclude loops in the communication topology. This means that, typically, we are considering leaderless coordination rather than a leader-follower approach. (Leaderless coordination is also considered, for example, in [34]–[38]). Finally, we allow for time-dependent communication patterns which are important if we want to take into account link failure and link creation, reconfigurable networks and nearest neighbor coupling.

The contribution of the paper is threefold. We present necessary and/or sufficient conditions on the communication

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topology guaranteeing convergence of the individual agents' states to a common value, thereby extending some of the results reported in [14], [27]–[29], and [31]. A second contribution concerns the analysis technique enabling us to obtain these results. We introduce a novel approach which is centered around the notion of convexity. Other studies that consider the connection between communication topology and system stability such as [14], [27]–[29], [31], [34], and [35] typically rely on algebraic graph theory, relating the graph topology with the algebraic structure of associated graph matrices. In this paper, instead of relying on algebraic graph theory, we propose a blend of graph-theoretic and system-theoretic tools to analyze stability. Our approach is of an inherently nonlinear nature with the notion of convexity playing a central role. (Independently of the present research, convexity ideas are also being applied to the distributed rendezvous problem in [20].) Third, the stability analysis is integrated within a formal framework of set-valued Lyapunov theory, which may be of independent interest.

The paper is organized as follows. Section II illustrates the main results of the paper by means of a linear example. Section III introduces the more general model which is studied in the remainder of the paper. Necessary and/or sufficient conditions for the convergence of the agents' states to a common value are presented in Section IV. Detailed proofs of these results are postponed until Section V. Section VI summarizes the main conclusions. In the paper, considerable attention is devoted to a precise formulation of the stability and convergence properties. For this purpose, we introduce in Appendix A several novel stability concepts, as well as a suitable Lyapunov characterization. Appendix B contains a graph-theoretic result that is used in the technical proofs of this paper. A preliminary version of this work has appeared as a conference paper [1].

Notation and terminology

We distinguish between the set-inclusion symbols $\subset$ and $\subseteq$. The symbol $\subset$ denotes strict inclusion whereas $\subseteq$ denotes nonstrict inclusion.

Let K be a subset of a finite-dimensional Euclidean space X. The set $\mathcal{B}(K,c)$ $(c>0)$ is defined as the set of points in X whose distance to K is strictly smaller than c. When K is a singleton $\{x\}$ the notation $\mathcal{B}(x,c)$ is used instead of $\mathcal{B}(\{x\},c)$.

Consider finite-dimensional Euclidean spaces X and Y. We denote by $\mathcal{P}(Y)$ the collection of all closed subsets of Y. A *set-valued function* $f: X \rightrightarrows Y$ is a (single-valued) function $f: X \to \mathcal{P}(Y)$. A set-valued function $f: X \rightrightarrows Y$ is called *upper semicontinuous* if for every $x \in X$ and every $\varepsilon > 0$ there is $\delta > 0$ such that $f(y) \in \mathcal{B}(f(x),\varepsilon)$ whenever $y \in \mathcal{B}(x,\delta)$.

The *relative interior* (ri) of a convex set C in finite-dimensional Euclidean space is the interior which results when this set is regarded as a subset of its affine hull and is denoted by $\mathrm{ri}(C)$. A *polytope* is a set which is the convex hull of finitely many points. See [39] for more information.

The set of nonnegative (respectively, strictly positive) real numbers is denoted by $\mathbb{R}_{\geq 0}$ (respectively, $\mathbb{R}_{>0}$).

II. Example

The present section illustrates the main results of the paper by means of a linear example. The results that we obtain for this example generalize results that have recently been reported in the literature. At the same time, the present section also introduces several notions that will be used repeatedly in the remainder of this paper (directed graph, neighbors, connectivity).

Definition 1 (Directed Graph): A directed graph is a pair $(\mathcal{N},\mathcal{A})$ where $\mathcal{N}$ is a nonempty, finite set and $\mathcal{A}$ is a subset of $\mathcal{N}\times\mathcal{N}$ satisfying $(k,k)\notin\mathcal{A}$ for all $k\in\mathcal{N}$. Elements of $\mathcal{N}$ are referred to as nodes and an element (k,l) of $\mathcal{A}$ is referred to as an arc from k to l.

Definition 2 (Neighbors): Consider a directed graph $(\mathcal{N},\mathcal{A})$ and a nonempty subset $\mathcal{L}\subseteq\mathcal{N}$. The set $\mathrm{Neighbors}(\mathcal{L},\mathcal{A})$ is the set of those nodes $k\in\mathcal{N}\setminus\mathcal{L}$ for which there is $l\in\mathcal{L}$ such that $(k,l)\in\mathcal{A}$. When $\mathcal{L}$ is a singleton $\{m\}$, the notation $\mathrm{Neighbors}(m,\mathcal{A})$ is used instead of $\mathrm{Neighbors}(\{m\},\mathcal{A})$.

Notice that any nonempty subset $\mathcal{L}\subseteq\mathcal{N}$ satisfies

$$\mathrm{Neighbors}\,(\mathcal{L},\mathcal{A}) = (\cup_{m\in\mathcal{L}}\,\mathrm{Neighbors}\,(m,\mathcal{A}))\setminus\mathcal{L}.$$

Definition 3 (Weighted Directed Graph): A weighted directed graph is a triple $(\mathcal{N},\mathcal{A},w)$ where $(\mathcal{N},\mathcal{A})$ is a directed graph and $w:\mathcal{A}\to\mathbb{R}_{>0}$ is a map associating to each arc (k,l) a strictly positive weight denoted by w_{kl}.

Remark 1: Notice that we have excluded self-loops in the definition of a directed graph. This is merely a matter of convention. It is important to emphasize that the stochastic matrix associated to a weighted directed graph according to (1) below does, in fact, have strictly positive diagonal elements.

Consider, for example, the weighted directed graph depicted in Fig. 1. Each node in this graph corresponds to an agent and each arc represents a communication channel. Each agent i $(i=1,\ldots,4)$ is being attributed a real state variable $x_i\in\mathbb{R}$.[1] Suppose that this weighted directed graph represents the communication pattern at time t. Agent 1 receives information from agent 2 and updates his state x_1 according to the weighted average

$$x_1(t+1) = \frac{1x_1(t)+\left(\frac{1}{2}\right)x_2(t)}{1+\frac{1}{2}}.$$

Notice that agent 1's own state $x_1(t)$ is taken into account with unit weight 1. Agent 2 receives information from agents 1 and 3 and updates his state x_2 according to the weighted average

$$x_2(t+1) = \frac{1x_1(t)+1x_2(t)+5x_3(t)}{1+1+5}.$$

Agents 3 and 4 do not receive any information

$$\begin{aligned} x_3(t+1) &= x_3(t)\\ x_4(t+1) &= x_4(t). \end{aligned}$$

We are interested in the dynamics of the x_i-variables that arises when the communication graph changes over time. In order to formulate the dynamics for general graphs, we associate to a weighted directed graph $\mathcal{G}=(\mathcal{N},\mathcal{A},w)$ with vertex set $\mathcal{N}=$

[1]Within the context of swarming models [11], [14], agent i could be a particle moving at constant speed in two-dimensional space and x_i could represent its heading. More general swarming models may be conceived, where x_i is not necessarily a scalar but possibly a vector representing the agent's configuration in space; see the following section.

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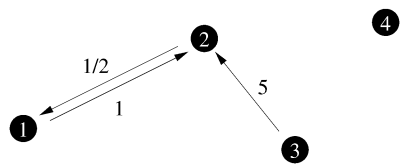

Fig. 1. Weighted directed graph.

$\{1,\ldots,n\}$ a real $n \times n$-matrix $A(\mathcal{G})$ whose components we define as shown in

$$(A(\mathcal{G}))_{kl}=\begin{cases} \dfrac{w_{lk}}{1+\sum\limits_{i\in\text{Neighbors}(k,\mathcal{A})} w_{ik}}, & \text{if } l\in\text{Neighbors}(k,\mathcal{A}) \\ \dfrac{1}{1+\sum\limits_{i\in\text{Neighbors}(k,\mathcal{A})} w_{ik}}, & \text{if } k=l \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Notice that the matrix $A(\mathcal{G})$ is *stochastic*; that is, it is a square matrix with nonnegative entries and with the property that its row sums are all equal to 1. In addition, its diagonal elements are strictly positive by construction. For the weighted directed graph of Fig. 1, for example, the matrix $A(\mathcal{G})$ becomes

$$\begin{pmatrix} \frac{2}{3} & \frac{1}{3} & 0 & 0 \\ \frac{1}{7} & \frac{1}{7} & \frac{5}{7} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Consider a sequence of weighted directed graphs $\mathcal{G}(t) = (\mathcal{N},\mathcal{A}(t),w(t))$ with common vertex set $\mathcal{N}=\{1,\ldots,n\}$ and where $t \in \mathbb{N}$. We are interested in the following discrete-time system on $\mathbb{R}^n$:

$$x(t+1) = A(\mathcal{G}(t))x(t). \tag{2}$$

In particular, we want to formulate necessary and/or sufficient conditions for the convergence of the n components $x_i(t)$ to a common value as $t \to \infty$. It should be clear that the convergence properties of (2) depend not only on the connectivity properties of the sequence of directed graphs but also on the associated weight functions. As we intend to focus on the role of the connectivity properties, we assume for simplicity that the weights are well-behaved by imposing uniform bounds on the weights. Here, we state two results that follow as a consequence of the general theory established in the paper (Theorems 2 and 3). The formulation of these results relies on the notion of connectivity.

Definition 4 (Connectivity): Consider a directed graph $(\mathcal{N},\mathcal{A})$. A node $k \in \mathcal{N}$ is connected to a node $l \in \mathcal{N}\setminus\{k\}$ if there is a path from k to l in the graph which respects the orientation of the arcs. Consider a sequence of directed graphs $(\mathcal{N},\mathcal{A}(t))$ with $t \in \mathbb{N}$. A node $k \in \mathcal{N}$ is connected to a node $l \in \mathcal{N}\setminus\{k\}$ across an interval $I \subseteq \mathbb{N}$ if k is connected to l for the directed graph $(\mathcal{N},\cup_{t\in I}\mathcal{A}(t))$.

Proposition 1 (Nonbidirectional): Consider a sequence of weighted directed graphs $\mathcal{G}(t) = (\mathcal{N},\mathcal{A}(t),w(t))$ with common vertex set $\mathcal{N}=\{1,\ldots,n\}$ and where $t \in \mathbb{N}$. Assume the existence of real numbers $e_{\max} \geq e_{\max} > 0$ such that $e_{\min} \leq w_{kl}(t) \leq e_{\max}$ for all $t \in \mathbb{N}$ and $(k,l) \in \mathcal{A}(t)$. If there is $T \geq 0$ such that for all $t_0 \in \mathbb{N}$ there is a node connected to all other nodes across $[t_0, t_0+T]$, then the n components $\zeta_i(t)$ of any solution ζ of (2) converge to a common value as $t \to \infty$.

Proposition 2 (Bidirectional): Consider a sequence of weighted directed graphs $\mathcal{G}(t) = (\mathcal{N},\mathcal{A}(t),w(t))$ with common vertex set $\mathcal{N}=\{1,\ldots,n\}$ and where $t \in \mathbb{N}$. Assume the existence of real numbers $e_{\max} \geq e_{\max} > 0$ such that $e_{\min} \leq w_{kl}(t) \leq e_{\max}$ for all $t \in \mathbb{N}$ and $(k,l) \in \mathcal{A}(t)$. Assume in addition that the graphs $(\mathcal{N},\mathcal{A}(t))$ are bidirectional[2] for all $t \in \mathbb{N}$. If for all $t_0 \in \mathbb{N}$ there is a node connected to all other nodes across $[t_0,\infty)$,[3] then the n components $\zeta_i(t)$ of any solution ζ of (2) converge to a common value as $t \to \infty$.

A. Discussion of Propositions 1 and 2

Let us mention some of the subtleties involved in the study of (2). First of all, (2) belongs to the class of linear time-varying systems, whose stability properties are hard to analyze in general. An eigenvalue analysis, for example, is no longer applicable if the system matrix depends explicitly on time. Second, the equilibrium value to which the components are shown to converge, is not known explicitly. This equilibrium value depends, in general, both on the initial data and on the sequence of weighted directed graphs. Third, it has been shown in [14] that, in general, there does not exist a time-invariant, quadratic Lyapunov function for (2). Partially motivated by this negative result we put forward in this paper a series of graph-theoretic and system-theoretic analysis tools which are *not* based on the linear structure of (2). As a consequence of our inherently nonlinear approach Propositions 1 and 2 hold *verbatim* if we replace the linear dynamics (2) by their nonlinear, saturated counterpart

$$x(t+1) = x(t) + \text{sat}(A(\mathcal{G}(t))x(t) - x(t))$$

where the saturation function sat should be interpreted componentwise as $\text{sat}(\cdot) = \max\{-1,\min\{\cdot,1\}\}$.

For the special case that the weights are all equal to one, (2) reduces to the model studied in [14, Ths. 1–2]. From a control perspective, an important contribution of [14] is that this work relates the information flow and communication structure with the stability properties of the group of agents. Apart from the observation that the present models are more general and possibly nonlinear, it is instructive to notice that we do not assume bidirectional communication (Proposition 1) and, when we consider bidirectional communication, we do not assume uniformity in the connectivity condition (Proposition 2).

By specializing Proposition 1 to the case of time-independent weighted directed graphs, we recover the following well-known graph theoretic result implicitly contained in [40, Secs. 7.2 and 7.3]. A stochastic $n \times n$-matrix A with positive diagonal elements has all but one of its eigenvalues strictly inside the unit circle (the only exception being the trivial eigenvalue 1) if (and only if—see Theorem 2) the associated directed graph $(\mathcal{N},\mathcal{A})$

$$\begin{aligned} \mathcal{N} &= \{1,\ldots,n\} \\ \mathcal{A} &= \{(k,l) \in \mathcal{N}\times\mathcal{N} : A_{lk} > 0 \text{ and } k \neq l\} \end{aligned}$$

[2]That is, $(k,l) \in \mathcal{A}(t)$ whenever $(l,k) \in \mathcal{A}(t)$. Notice, however, that we do not require the weights $w_{kl}(t)$ and $w_{lk}(t)$ to be equal.

[3]Since the graphs are assumed to be bidirectional this means that all nodes are connected to all other nodes across $[t_0,\infty)$.

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has a node connected to all other nodes. Indeed, observe that the right eigenvector $(1, \ldots, 1)$ corresponding to the trivial eigenvalue 1 of a stochastic matrix determines the equilibrium set $\{x \in \mathbb{R}^n : x_1 = \cdots = x_n\}$ of (2). The convergence property established in Proposition 1 corresponds to all other eigenvalues being located strictly inside the unit circle. As mentioned before, we recover this well-known graph theoretic result using inherently nonlinear analysis tools. Moreover, the present approach enables us to go beyond this result and consider time-dependent stochastic matrices.

As a preview of the remainder of the paper, let us announce that our approach is based on the convexity property or contraction property of (2), which says that the future, updated value of any agent in the network is a convex combination of its current value as well as the current values of its neighbors. It is thus clear that $\max\{x_1, \ldots, x_n\}$ is a nonincreasing function of time. Likewise $\min\{x_1, \ldots, x_n\}$ cannot decrease. In the remainder of this paper, we will work out these ideas in a more general setting and study their consequences for the stability properties of multi-agent systems with time-dependent communication links. The convexity or contraction property mentioned here has actually been known for a long time, notably in the context of nonnegative and stochastic matrices. It serves as a starting point for the ergodicity analysis of products of stochastic matrices in [32]. See also [45], where the relationship between ergodicity theory and distributed algorithms is further explored.

III. Multiagent Dynamics

We introduce here the dynamical equations that will be studied in the remainder of this paper. These equations model a network of n interacting agents. The individual agents share a common state space X which is assumed to be finite-dimensional Euclidean. As a model for this multiagent network we consider a nonlinear discrete-time system on X^n

$$x(t+1) = f(t, x(t)) \tag{3}$$

or, expressed in terms of the individual agents' states

$$\begin{aligned} x_1(t+1) &= f_1(t, x_1(t), \ldots, x_n(t)) \\ &\vdots \\ x_n(t+1) &= f_n(t, x_1(t), \ldots, x_n(t)). \end{aligned}$$

Notice that the individual x_k-variables need not be scalars; they may belong to a common Euclidean space of arbitrary finite dimension. For simplicity of the presentation we assume that the map $f : \mathbb{N} \times X^n \to X^n$ is continuous.

The formulation in (3) is still very general and does not yet capture the constraints imposed by the limited interagent communication. In order to incorporate the limited interagent communication in the model, it is common to introduce a sequence of directed graphs $(\mathcal{N}, \mathcal{A}(t))$ with common node set $\mathcal{N} = \{1, \ldots, n\}$ and with $t \in \mathbb{N}$. The directed graph $(\mathcal{N}, \mathcal{A}(t))$ characterizes the communication links at time t. Each node of this graph corresponds to an agent and each arc represents a communication channel. The communication graph $(\mathcal{N}, \mathcal{A}(t))$ determines what information is available for which agent at time t. The following assumption relates the sequence of communication graphs $(\mathcal{N}, \mathcal{A}(t))$ with the discrete-time dynamics (3). It is a strict convexity assumption requiring that, for each agent k, the updated state $x_k(t+1)$ is a strict convex combination of agent k's current state $x_k(t)$ and the current states $x_i(t)$ of its neighbors $i \in \mathrm{Neighbors}(k, \mathcal{A}(t))$.

Assumption 1 (Convexity): Associated to each directed graph $(\mathcal{N}, \mathcal{A})$ with node set $\mathcal{N} = \{1, \ldots, n\}$, each agent k and each state $x \in X^n$, there is a compact set $e_k(\mathcal{A})(x) \subset X$ satisfying

1) $$f_k(t, x) \in e_k(\mathcal{A}(t))(x) \qquad \forall\, t \in \mathbb{N} \qquad \forall\, x \in X^n; \tag{4}$$
2) $e_k(\mathcal{A})(x_1, \ldots, x_n) = \{x_k\}$ whenever the states of agent k and agents $i \in \mathrm{Neighbors}(k, \mathcal{A})$ are all equal;
3) $e_k(\mathcal{A})(x_1, \ldots, x_n)$ is contained in the relative interior of the convex hull of the states of agent k and agents $i \in \mathrm{Neighbors}(k, \mathcal{A})$ whenever the states of agent k and agents $i \in \mathrm{Neighbors}(k, \mathcal{A})$ are not all equal;
4) $e_k(\mathcal{A})(x)$ depends continuously on x; that is, the set-valued function $e_k(\mathcal{A}) : X^n \rightrightarrows X$ is continuous.

The following examples illustrate that the present model finds wide application in a variety of fields including synchronization, swarming and distributed decision making. The necessary and/or sufficient conditions for the convergence of the individual agents' states to a common value that are presented in the following section apply directly to these examples.

Example 1 (Linear): The linear model of Section II falls within the scope of the present study. Consider a sequence of weighted directed graphs $\mathcal{G}(t) = (\mathcal{N}, \mathcal{A}(t), w(t))$ with common vertex set $\mathcal{N} = \{1, \ldots, n\}$ and where $t \in \mathbb{N}$. Assume the existence of real numbers $e_{\max} \geq e_{\max} > 0$ such that the weights $w_{kl}(t)$ satisfy $e_{\min} \leq w_{kl}(t) \leq e_{\max}$ for all $t \in \mathbb{N}$ and $(k, l) \in \mathcal{A}(t)$. The linear model studied in Section II corresponds to the case of $X = \mathbb{R}$ and $f(t, x) = A(\mathcal{G}(t))x$ with the matrix $A(\mathcal{G})$ defined in (1). The discrete-time map f satisfies Assumption 1 with $e(\mathcal{A})(x)$ being the set of all possible values of $A(\mathcal{G})x$ that are obtained by considering all possible weighted directed graphs $\mathcal{G}$ with the given arc set $\mathcal{A}$ and with weights contained in the compact interval $[e_{\min}, e_{\max}]$.

Example 2 (Synchronization): Consider a population of oscillators that share a common state space S^1. Denote the state of oscillator k by θ_k and consider for each communication graph $(\mathcal{N}, \mathcal{A})$ the Kuramoto equation [8], [9]

$$\dot{\theta}_k = \omega_k + \sum_{i \in \mathrm{Neighbors}(k, \mathcal{A})} a_{ik} \sin(\theta_i - \theta_k), \quad k = 1, \ldots, n \tag{5}$$

with strictly positive coupling strengths $a_{ik} > 0$. Let us consider the case that the ω_k are all equal. In this case, we may assume without loss of generality (by introducing a rotating reference frame) that $\omega_1 = \cdots = \omega_n = 0$. We restrict attention to angles in the interval $(-\pi/2, \pi/2)$ and introduce local coordinates

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$(-\pi/2, \pi/2) \to \mathbb{R} : \theta_k \mapsto x_k = \tan(\theta_k)$. Expressed in terms of these local coordinates, the Kuramoto equation becomes[4]

$$\dot{x}_k = \sum_{i \in \mathrm{Neighbors}(k,\mathcal{A})} a_{ik} \frac{\sqrt{1+x_k^2}}{\sqrt{1+x_i^2}}(x_i - x_k),\ k = 1, \ldots, n. \tag{6}$$

The continuous-time system (6) may be studied within the present framework by introducing its time-1 map $\psi(\mathcal{A}) : \mathbb{R}^n \to \mathbb{R}^n$ which depends, of course, on the communication structure $\mathcal{A}$. Explicitly, consider a time-sequence $(\mathcal{N}, \mathcal{A}(t))$ $(t \in \mathbb{N})$ and let $f(t,x) = \psi(\mathcal{A}(t))(x)$. This discrete-time map f satisfies Assumption 1 with $e(\mathcal{A})(x) = \{\psi(\mathcal{A})(x)\}$. The present paper allows to study synchronization of oscillators with time-varying and nonbidirectional coupling.

Example 3 (Consensus Algorithm): The following nonlinear consensus protocol is studied in [27]. Let $X = \mathbb{R}$ and consider for a given communication graph $(\mathcal{N}, \mathcal{A})$ the continuous-time system on $\mathbb{R}^n$

$$\dot{x}_k = \sum_{i \in \mathrm{Neighbors}(k,\mathcal{A})} \gamma_{ik}(x_i - x_k), \qquad k = 1, \ldots, n \tag{7}$$

where the functions $\gamma_{ik} : \mathbb{R} \to \mathbb{R}$ are uneven, locally Lipschitz and strictly increasing for all $(i,k) \in \mathcal{A}$. Similarly as before (Example 2), the continuous-time system (7), as well as time-dependent versions of it, may be studied within the present framework by introducing its time-1 map $\psi(\mathcal{A}) : \mathbb{R}^n \to \mathbb{R}^n$. This paper allows to study nonlinear consensus protocols with arbitrary time-varying and nonbidirectional communication topologies.

Example 4 (Swarming): Reference [11] proposes a simple model to investigate self-ordered motion in systems of particles with biologically motivated interaction. The model consists of a collection of particles moving at constant speed. At each time step a given particle assumes the average direction of motion of the particles in its neighborhood with some random perturbation added. Denote the direction of motion of particle k by $\theta_k \in S^1$. The following model is studied in [11]:

$$\theta_k(t+1) = \arctan\left(\frac{\sum_{i \in \{k\} \cup \mathrm{Neighbors}(k,\mathcal{A}(t))} \sin(\theta_i(t))}{\sum_{i \in \{k\} \cup \mathrm{Neighbors}(k,\mathcal{A}(t))} \cos(\theta_i(t))}\right), \quad k = 1, \ldots, n \tag{8}$$

where we have omitted an additive disturbance term. Here, $\mathcal{A}(t)$ characterizes the nearest neighbor coupling at time t. The nonlinear swarming model (8) may be analyzed with the tools of this paper by restricting attention to angles in the interval $(-\pi/2, \pi/2)$ and introducing local coordinates $(-\pi/2, \pi/2) \to \mathbb{R} : \theta_k \mapsto x_k = \tan(\theta_k)$, similarly as in Example 2.

[4]The reason for this coordinate transformation is simply that, in the present study, we have chosen to assume that the common state–space of the individual agents is an unbounded Euclidean space (and not an open subset of Euclidean space). Also, the formulation of the stability notions in Appendix A assumes that the state–space is unbounded Euclidean. For these reasons, we perform a preliminary coordinate transformation from $(-\pi/2, \pi/2)$ to $\mathbb{R}$.

If the headings θ_k are close to each other, then (8) may be approximated by the linear equation

$$x_k(t+1) = \frac{\sum_{i \in \{k\} \cup \mathrm{Neighbors}(k,\mathcal{A}(t))} x_i(t)}{1 + \#\mathrm{Neighbors}(k, \mathcal{A}(t))}, \qquad k = 1, \ldots, n \tag{9}$$

with $x_k \in \mathbb{R}$ and where $\#\mathrm{Neighbors}(k, \mathcal{A}(t))$ denotes the number of neighbors of agent k at time t. This is the swarming model studied in [14] and corresponds to the linear example of Section II with all weights equal to 1.

Assumption 1 plays a central role in the forthcoming analysis. Its importance is already reflected in the following simple but appealing result.

Lemma 1: Consider a time sequence of directed graphs $(\mathcal{N}, \mathcal{A}(t))$ with common node set $\mathcal{N} = \{1, \ldots, n\}$ and a continuous update map $f : \mathbb{N} \times X^n \to X^n$ satisfying Assumption 1. The convex hull of the individual agents' states, denoted by $\widetilde{V}(x) = \mathrm{conv}\{x_1, \ldots, x_n\} \subset X$, does not grow along the solutions of the discrete-time system (3); that is

$$\widetilde{V}(f(t,x)) \subseteq \widetilde{V}(x) \qquad \forall\, t \in \mathbb{N} \qquad \forall\, x \in X^n. \tag{10}$$

Proof: Assumption 1 implies

$$f_k(t,x) \in e_k(\mathcal{A}(t))(x) \subseteq \mathrm{conv}\{x_1, \ldots, x_n\} \quad \forall\, k \in \{1, \ldots, n\} \qquad \forall\, t \in \mathbb{N} \qquad \forall\, x \in X^n \tag{11}$$

from which (10) follows immediately. □

Within the current framework, the statement of Lemma 1 may seem almost trivial; it arises as a straightforward consequence of Assumption 1. However, its nontrivial contribution lies precisely in the observation that many examples reported in the literature satisfy Assumption 1 and, hence, may be studied by means of Lemma 1. The convex hull of the individual agents' states serves as a measure for disagreement. Lemma 1 states that, in terms of this measure, the level of disagreement cannot increase with time. A consensus is reached when the convex hull reduces to a singleton, as this implies a common state value for all agents. In order for this to happen, additional assumptions need to be imposed. In the forthcoming sections, we will present necessary and/or sufficient conditions ensuring that the convex hull indeed approaches a singleton.

We end this section with some remarks on the limitations of the present model.

Remark 2 (Nonstrict Convexity): Assumption 1 is a strict convexity assumption. For some applications it may be desirable to relax this assumption. For example, the discrete-time system

$$x_k(t+1) = \max_{i \in \{k\} \cup \mathrm{Neighbors}(k,\mathcal{A}(t))} x_i(t), \qquad k = 1, \ldots, n \tag{12}$$

does not satisfy Assumption 1, but satisfies instead a nonstrict convexity assumption: $x_k(t+1)$ belongs to (the boundary of) the convex hull of the state $x_k(t)$ and the states $x_i(t)$ of agents $i \in \mathrm{Neighbors}(k, \mathcal{A}(t))$. The study of dynamical equations satisfying a nonstrict convexity assumption is beyond the scope of this paper.

Remark 3 (Non-Euclidean Space): By assuming that the common state space for the individual agents is Euclidean and by imposing a convexity condition on the dynamics, we exclude some global phenomena arising, for example, in the

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study of synchronization of periodic motions [8] and in attitude alignment problems [13]. Indeed, the natural state–space for the individual agents in these examples is, respectively, S^1 and $SO(3)$ and the global features of the dynamics that are related to the nontrivial topology of these manifolds, fall outside the scope of the present method. Nevertheless, local results may be obtained within the present framework, for example, by introducing suitable coordinate charts. This has already been illustrated in Examples 2 and 4.

IV. Convergence Results

Before actually studying the convergence properties, we first present the following stability and boundedness result, which is included here for reasons of completeness. It is a formalization of the observation that the level of disagreement can not increase with time.

Theorem 1: Consider a time sequence of directed graphs $(\mathcal{N}, \mathcal{A}(t))$ with common node set $\mathcal{N} = \{1, \ldots, n\}$ and a continuous update map $f : \mathbb{N} \times X^n \to X^n$ satisfying Assumption 1. The discrete-time system (3) is uniformly stable and uniformly bounded with respect to the collection of equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant.

For a precise definition of the notions of *uniform stability* and *uniform boundedness with respect to a collection of equilibrium solutions* we refer to Appendix A. Theorem 1 is a consequence of Lemma 1. A formal proof is provided in Section V.

It is important to emphasize that Theorem 1 leaves open the question of convergence, which is our primary object of interest. Contrary to what might be expected from Lemma 1 or Theorem 1, the dynamics of (3) may be surprisingly complex. For example, the agents' states may fail to converge to a common value, even in the presence of persistent interagent communication; see Section IV-C. In the remainder of the paper a blend of graph-theoretic and system-theoretic tools will be used in order to establish necessary and/or sufficient conditions for convergence (attractivity).

A. Uniform Global Attractivity

We first restrict attention to global attractivity uniform with respect to initial time. We provide a necessary and sufficient condition for uniform global attractivity of (3). The condition that we present does not involve the actual discrete-time map f; it only involves the sequence of communication graphs $(\mathcal{N}, \mathcal{A}(t))$.

Theorem 2 (Uniform Global Attractivity): Consider a time sequence of directed graphs $(\mathcal{N}, \mathcal{A}(t))$ with common node set $\mathcal{N} = \{1, \ldots, n\}$ and a continuous update map $f : \mathbb{N} \times X^n \to X^n$ satisfying Assumption 1. The discrete-time system (3) is uniformly globally attractive with respect to the collection of equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant if and only if there is $T \geq 0$ such that for all $t_0 \in \mathbb{N}$ there is a node connected to all other nodes across $[t_0, t_0 + T]$.

For a precise definition of the notion of *uniform global attractivity with respect to a collection of equilibrium solutions* we refer to Appendix A. The proof of the *if*-part of Theorem 2 is based on the observation that the convex hull of the individual agents' states, although not necessarily decreasing at each time step, is in fact decreasing over time intervals of sufficient length. The proof of the *only if*-part is based upon the graph theoretic result from Appendix B. A formal proof is provided in Section V.

B. Nonuniform Global Attractivity

The previous subsection has presented a necessary and sufficient condition for uniform global attractivity. Not surprisingly, this condition involves a connectivity requirement on the sequence of directed graphs, uniform with respect to initial time. We now turn attention to global attractivity, not necessarily uniform with respect to initial time. Unlike the previous subsection where the situation is quite clear, the study of nonuniform attractivity turns out to be much more subtle. We start with a necessary condition for global attractivity, not necessarily uniform with respect to initial time.

Proposition 3 (Global Attractivity): Consider a time sequence of directed graphs $(\mathcal{N}, \mathcal{A}(t))$ with common node set $\mathcal{N} = \{1, \ldots, n\}$ and a continuous update map $f : \mathbb{N} \times X^n \to X^n$ satisfying Assumption 1. Global attractivity of the discrete-time system (3) with respect to the collection of equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant implies for all $t_0 \in \mathbb{N}$ the existence of a node connected to all other nodes across $[t_0, \infty)$.

For a precise definition of the notion of (nonuniform) *global attractivity with respect to a collection of equilibrium solutions* we refer to Appendix A. The proof of Proposition 3 is very similar to the proof of the *only if*-part of Theorem 2 and is omitted. The necessary condition featuring in Proposition 3 may be interpreted as a nonuniform version of the necessary and sufficient condition featuring in Theorem 2. It may therefore seem tempting to conjecture that this necessary condition is also sufficient for global attractivity, not necessarily uniform with respect to time. However, the following counterexample shows that this is not the case.

C. Counterexample

The counterexample is concerned with three agents sharing a common state–space $X = \mathbb{R}$. Among all possible directed graphs on the vertex set $\{1, 2, 3\}$ we consider

$$\begin{aligned} \mathcal{A}_a &= \{(1,2)\} \\ \mathcal{A}_b &= \{(1,2),(2,1)\} \\ \mathcal{A}_c &= \{(3,2)\} \\ \mathcal{A}_d &= \{(2,3),(3,2)\}. \end{aligned} \tag{13}$$

We will introduce a sequence of directed graphs which consists of the concatenation of finite sequences of the form B_s ($s \in \{0\} \cup \mathbb{N}$)

$$B_s = \underbrace{\mathcal{A}_a, \ldots, \mathcal{A}_a}_{2s}, \mathcal{A}_b, \underbrace{\mathcal{A}_c, \ldots, \mathcal{A}_c}_{2s+1}, \mathcal{A}_d.$$

Proposition 4: Let the number of agents be given by $n = 3$ and consider the sequence of communication graphs $(\mathcal{N}, \mathcal{A}(t))$ with $t \in \mathbb{N}$ corresponding to the concatenation $B_0, B_1, B_2, B_3, \ldots$ Let the common state space for the individual agents be given by $X = \mathbb{R}$ and consider the discrete-time map f corresponding to the linear example of Section II with all the weights equal to one. Let ζ be the solution of (3) with

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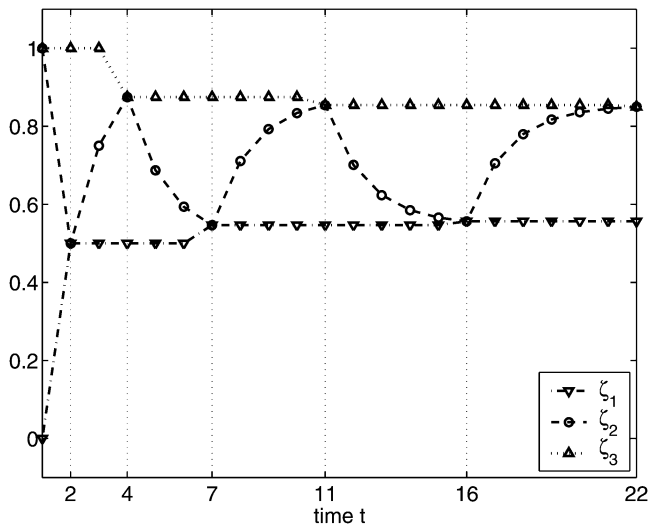

Fig. 2. Individual agents' states fail to converge to a common value in spite of persistent interagent communication.

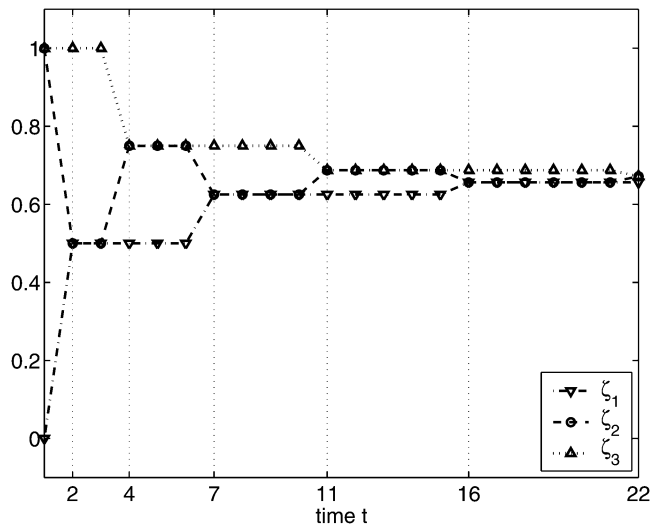

Fig. 3. Convergence of the individual agents' states to a common value is obtained by reducing the communication.

initial data $\zeta_1(1) = 0$ and $\zeta_2(1) = \zeta_3(1) = 1$. Then, the three components of $\zeta(t)$ do not converge to a common value as $t \to \infty$.

Proof: In order to show that the three components of $\zeta(t)$ do not converge to a common value as $t \to \infty$, we evaluate the difference $\zeta_3 - \zeta_1$ at the sequence of time-instants $t_1, t_2, t_3, \ldots \to \infty$ determined by $t_{p+1} - t_p = p + 1$ for all $p \in \mathbb{N}$ and $t_1 = 2$ (Fig. 2). For the ease of notation, let us denote $\zeta_3(t_p) - \zeta_1(t_p)$ by $v(p)$. It is not difficult to see that

$$v(p+1) = v(p) - \frac{1}{2^{p+1}} v(p) = \frac{2^{p+1} - 1}{2^{p+1}} v(p) \quad \forall\, p \in \mathbb{N} \tag{14}$$

$$v(1) = \frac{1}{2} \tag{15}$$

from which it follows, among others, that $0 < v(p) < 1$ for all $p \in \mathbb{N}$. Clearly, $v(p)$ is decreasing with p and converges to a limit as $p \to \infty$. The total accumulative decrease of v satisfies

$$\sum_{p=1}^{\infty} (v(p) - v(p+1)) = \sum_{p=1}^{\infty} \frac{1}{2^{p+1}} v(p) < \sum_{p=1}^{\infty} \frac{1}{2^{p+1}} = \frac{1}{2} \tag{16}$$

where we have used the recursive relation (14) and the observation that $0 < v(p) < 1$ for all $p \in \mathbb{N}$. Since $v(1) = 1/2$, we conclude that

$$\begin{aligned} \limsup_{p \to \infty} \zeta_3(t_p) - \zeta_1(t_p) &\geq \lim_{p \to \infty} v(p) \\ &= v(1) - \sum_{p=1}^{\infty} (v(p) - v(p+1)) \\ &> 0. \end{aligned} \tag{17}$$

□

We end this subsection with a discussion of the counterexample. First of all, notice that the sequence of directed graphs featuring in the counterexample satisfies the necessary condition of Proposition 3. The union of the arc sets $\mathcal{A}(t)$ over any interval of the form $[t_0, \infty)$ is given by $\{(1,2), (2,1), (2,3), (3,2)\}$ which actually corresponds to a connected, bidirectional graph. This counterexample thus clearly shows that the necessary condition of Proposition 3 is not sufficient for global attractivity. In other words, persistent interagent communication does not necessarily guarantee convergence of the individual agents' states to a common value.

The counterexample points toward an interesting phenomenon, namely that more information exchange does not necessarily lead to improved convergence properties, and may eventually even lead to a loss of convergence. To be more precise, if the arc sets $\mathcal{A}_c$ and $\mathcal{A}_c$ are replaced by empty sets, thus effectively reducing the communication, then the discrete-time system featuring in Proposition 4 becomes globally attractive. This follows from Theorem 3 stated below and is illustrated in Fig. 3. By taking $\mathcal{A}_c$ and $\mathcal{A}_c$ as defined in (13) instead of empty sets, thus effectively increasing the communication, the agents' states fail to converge to a common value. In the present example the loss of convergence may be understood as follows. Agent 2 serves as an intermediate agent facilitating communication between agents 1 and 3. In other words, agent 2 is responsible for carrying information from agent 3 to agent 1 and *vice versa*. However, the negative effect of the unidirectional communication links, say link (1,2) in $\mathcal{A}_a$, is to (partially) replace agent 3's information with agent 1's information before it is passed to agent 1. Notice that the loss of convergence observed in the counterexample can only occur in the context of time-dependent and nonbidirectional communication; see Theorems 2 and 3.

D. Bidirectional Communication

The study of global attractivity of (3), not necessarily uniform with respect to initial time, is simplified considerably when bidirectional communication is assumed.

Theorem 3 (Bidirectional): Consider a time sequence of directed graphs $(\mathcal{N}, \mathcal{A}(t))$ with common node set $\mathcal{N} = \{1, \ldots, n\}$ and a continuous update map $f : \mathbb{N} \times X^n \to X^n$ satisfying Assumption 1. Assume in addition that the communication graphs $(\mathcal{N}, \mathcal{A}(t))$ are bidirectional for all $t \in \mathbb{N}$. The discrete-time system (3) is globally attractive with respect to the collection of equilibrium solutions

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$x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant if and only if for all $t_0 \in \mathbb{N}$ there is a node connected to all other nodes across $[t_0, \infty)$.[5]

The proof of the *if*-part of Theorem 3 is based on an analysis of ω-limit sets. It is, of course, well-known that ω-limit sets play a central role in the stability analysis of dynamical systems through the celebrated LaSalle principle, but it is quite remarkable that ω-limit sets prove to be useful in the present context, since we are considering time-varying difference equations, not necessarily periodic. A proof of Theorem 3 is provided in Section V. Reference [41] provides further insight why bidirectional communication simplifies the analysis and enables nonuniform attractivity results.

V. Technical Proofs

A. Proof of Theorem 1

We provide a formal proof based upon Theorem 4 of Appendix A. In order to apply Theorem 4 we introduce the set-valued function $V : X^n \rightrightarrows X^n$ according to

$$V(x_1, \ldots, x_n) = (\mathrm{conv}\{x_1, \ldots, x_n\})^n \qquad \forall\, x \in X^n.$$

The set-valued function V is derived from $\widetilde{V}$ featuring in Lemma 1 following the procedure of Remark 6 in Appendix A with $\mathcal{P}(X), \subset$ playing the role of $Z, \prec$. The set-valued function V is easily seen to be globally Lipschitz. Theorem 1 follows immediately from Theorem 4 and Lemma 1. □

B. Proof of Theorem 2

(Only if, proof by contraposition.) Assume that for every $T \geq 0$ there is $t_0 \in \mathbb{N}$ such that no node is connected to all other nodes across $[t_0, t_0 + T]$. By Theorem 5 of Appendix B, this implies that for every $T \geq 0$ there is $t_0 \in \mathbb{N}$ and there are nonempty, disjoint subsets $\mathcal{L}_1, \mathcal{L}_2 \subset \mathcal{N}$ such that $\mathrm{Neighbors}(\mathcal{L}_1, \mathcal{A}(t))$ and $\mathrm{Neighbors}(\mathcal{L}_2, \mathcal{A}(t))$ are both empty for all $t \in [t_0, t_0 + T]$. Pick two different elements $y, \overline{y} \in X$ and consider any solution ζ of (3) with initial data

$$\zeta_j(t_0) \begin{cases} = y & \forall\, j \in \mathcal{L}_1 \\ = \overline{y} & \forall\, j \in \mathcal{L}_2 \\ \in \mathrm{conv}\{y, \overline{y}\} & \forall\, j \in \mathcal{N} \setminus (\mathcal{L}_1 \cup \mathcal{L}_2). \end{cases} \tag{18}$$

Assumption 1 implies that, at time $t_0 + T + 1$, we still have

$$\zeta_j(t_0 + T + 1) \begin{cases} = y & \forall\, j \in \mathcal{L}_1 \\ = \overline{y} & \forall\, j \in \mathcal{L}_2 \\ \in \mathrm{conv}\{y, \overline{y}\} & \forall\, j \in \mathcal{N} \setminus (\mathcal{L}_1 \cup \mathcal{L}_2) \end{cases} \tag{19}$$

since $\mathrm{Neighbors}(\mathcal{L}_1, \mathcal{A}(t))$ and $\mathrm{Neighbors}(\mathcal{L}_2, \mathcal{A}(t))$ are both empty for all $t \in [t_0, t_0 + T]$. As the time T may be chosen arbitrarily large, it is not difficult to see that this contradicts uniform global attractivity of (3) with respect to the equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant.

(If) Let $T \geq 0$ be such that for all $t_0 \in \mathbb{N}$ there is a node connected to all other nodes across $[t_0, t_0 + T]$. Consider an arbitrary solution ζ of (3) and an arbitrary time $t_0 \in \mathbb{N}$ and assume that the $\zeta_j(t_0)$'s ($j \in \mathcal{N}$) are not all equal. We first show that $\widetilde{V}(\zeta(t_0 + (n-1)(T+1)))$ is strictly contained in $\widetilde{V}(\zeta(t_0))$, where $\widetilde{V}$ is the set-valued function introduced in Lemma 1: $\widetilde{V}(x) = \mathrm{conv}\{x_1, \ldots, x_n\}$.

In order to show this, we introduce a number of auxiliary functions. Denote the m vertices ($2 \leq m \leq n$) of the polytope $\widetilde{V}(\zeta(t_0))$ by $y_1, \ldots, y_m \in X$. Associate to each vertex y_i the set-valued function $a_i : [t_0, \infty) \rightrightarrows \mathcal{N}$ identifying the agents located at that vertex

$$a_i(t) = \{j \in \mathcal{N} : \zeta_j(t) = y_i\} \qquad \forall\, t \in [t_0, \infty) \qquad \forall\, i = 1, \ldots, m. \tag{20}$$

The strict convexity assumption (Assumption 1) implies that an agent which is not located at a vertex y_i at some time $t \geq t_0$ can never reach this vertex in finite time

$$a_i(t+1) \subseteq a_i(t) \qquad \forall\, t \in [t_0, \infty) \qquad \forall\, i = 1, \ldots, m. \tag{21}$$

We now establish a strict decrease property for the functions a_i. We show that, under the connectivity assumption of the theorem, at least $m - 1$ of the functions $a_1, \ldots, a_m$ are empty-valued at $t = t_0 + (n-1)(T+1)$. Consider any intermediate time interval $[t_0 + p(T+1), t_0 + p(T+1) + T]$ ($p = 0, \ldots, n-2$). The assumption of the theorem guarantees the existence of an agent k which is connected to all other agents across $[t_0 + p(T+1), t_0 + p(T+1) + T]$. We distinguish two cases.

- If this agent k is located in one of the vertices at time $t_0 + p(T+1)$; that is, if there is $i \in \{1, \ldots, m\}$ such that $k \in a_i(t_0 + p(T+1))$, then all the other vertices $j \in \{1, \ldots, m\} \setminus \{i\}$ where agents are located at time $t_0 + p(T+1)$, that is, $a_j(t_0 + p(T+1)) \neq \emptyset$, satisfy the following: $\mathrm{Neighbors}(a_j(t_0 + p(T+1)), \mathcal{A}(t)) \neq \emptyset$ for some $t \in [t_0 + p(T+1), t_0 + p(T+1) + T]$ and hence, by the strict convexity assumption (Assumption 1), $a_j(t_0 + (p+1)(T+1))$ is strictly contained in $a_j(t_0 + p(T+1))$.
- If this agent k is not located in one of the vertices at time $t_0 + p(T+1)$; that is, if $k \notin a_j(t_0 + p(T+1))$ for all $j \in \{1, \ldots, m\}$, then all vertices $j \in \{1, \ldots, m\}$ where agents are located at time $t_0 + p(T+1)$, that is, $a_j(t_0 + p(T+1)) \neq \emptyset$, satisfy the following: $\mathrm{Neighbors}(a_j(t_0 + p(T+1)), \mathcal{A}(t)) \neq \emptyset$ for some $t \in [t_0 + p(T+1), t_0 + p(T+1) + T]$ and hence, by the strict convexity assumption (Assumption 1), $a_j(t_0 + (p+1)(T+1))$ is strictly contained in $a_j(t_0 + p(T+1))$.

Since the aforementioned argument holds for every time interval $[t_0 + p(T+1), t_0 + p(T+1) + T]$ ($p = 0, \ldots, n-2$) and since the maximum number of agents at each vertex at time t_0 is not greater than $n - 1$, we may conclude that at least $m - 1$ of the functions $a_1, \ldots, a_m$ are empty-valued at $t = t_0 + (n-1)(T+1)$. This establishes a strong decrease property for the set-valued function $\widetilde{V}$. The polytope $\widetilde{V}(\zeta(t_0 + (n-1)(T+1)))$ is strictly contained in the polytope $\widetilde{V}(\zeta(t_0))$ and both sets have not more than one vertex in common.

The remainder of the proof is based upon Theorem 4 of Appendix A. In order to prepare for an application of Theorem 4,

[5]Since the graphs are assumed to be bidirectional this means that all nodes are connected to all other nodes across $[t_0, \infty)$.

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we introduce the set-valued function $V : X^n \rightrightarrows X^n$ according to

$$V(x_1,\ldots,x_n) = (\mathrm{conv}\{x_1,\ldots,x_n\})^n \qquad \forall\, t \in \mathbb{N} \qquad \forall\, x \in X^n$$

similarly as in the proof of Theorem 1. In addition we also introduce a real-valued function $\beta : X^n \to \mathbb{R}$ according to

$$\beta(x) = \inf_{\zeta_0,\ldots,\zeta_\tau} \mu(V(\zeta_0)) - \mu(V(\zeta_\tau)), \qquad \tau = (n-1)(T+1) \tag{22}$$

where μ denotes the diameter of a set and where the infimum is taken over all sequences $\zeta_0,\ldots,\zeta_\tau$ in X^n satisfying

$$\begin{aligned} &\zeta_0 = x \\ &\zeta_1 \in e(\mathcal{A}(t_0))(\zeta_0) \\ &\quad\vdots \\ &\zeta_\tau \in e(\mathcal{A}(t_0+\tau-1))(\zeta_{\tau-1}) \end{aligned} \tag{23}$$

for some $t_0 \in \mathbb{N}$. The ζ_i may be interpreted as the states that are possibly reachable from x in i time steps according to Assumption 1.

We establish some useful properties of β. First, the collection of all sequences $\zeta_0,\ldots,\zeta_\tau$ in X^n satisfying (23) for some $t_0 \in \mathbb{N}$

- is nonempty and compact for all $x \in X^n$;
- depends continuously on x.

This follows from the observations that the set-valued functions $e(\mathcal{A}) : X^n \rightrightarrows X^n$ are continuous and take nonempty, compact values (implied by Assumption 1) and that there is only a finite number of possible sequences of τ communication graphs. Second, the expression $\mu(V(\zeta_0)) - \mu(V(\zeta_\tau))$ being minimized in (22)

- depends continuously on ζ_0 and ζ_τ: indeed, this follows from the global Lipschitz property of V;
- is zero whenever the n components of x are all equal: indeed, in this case the only sequence satisfying (23) is $\zeta_0 = \cdots = \zeta_\tau = x$;
- is strictly positive whenever the n components of x are not all equal: indeed, this follows from the strict decrease property of $\widetilde{V}$ that has been established above and the relation $V(x) = (\widetilde{V}(x))^n$.

Putting all the elements together, we conclude that the function $\beta : X^n \to \mathbb{R}$ is continuous and positive definite with respect to $\{(x_1,\ldots,x_n) \in X^n : x_1 = \cdots = x_n\}$.

The proof is concluded with an application of Theorem 4 (see Remark 5) which establishes uniform global asymptotic stability of (3). □

C. Proof of Theorem 3

(Only if) This follows from Proposition 3.

(If) It suffices to prove that every solution of (3) converges to one of the equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant. Indeed, by continuity and compactness arguments, convergence of all individual solutions actually implies global attractivity in the sense of Definition 5 of Appendix A.[6]

In the remainder of the proof we show that every solution of (3) converges to one of the equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant. Consider arbitrary $t_0 \in \mathbb{N}$ and $x_0 \in X^n$, and let ζ denote the solution of (3) with initial data $\zeta(t_0) = x_0$. Let Ω denote the ω-limit set of ζ. We start with observing that

$$\widetilde{V}(x) = \widetilde{V}(\overline{x}) \qquad \forall\, x, \overline{x} \in \Omega \tag{24}$$

where $\widetilde{V}$ is the set-valued function introduced in Lemma 1. Indeed, the existence of $x, \overline{x} \in \Omega$ with $\widetilde{V}(x) \neq \widetilde{V}(\overline{x})$ would contradict the nonincrease property established in Lemma 1. We denote the constant value of $\widetilde{V}$ on Ω by $\widetilde{V}(\Omega)$.

Clearly, in order to establish that ζ converges to one of the equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant, it suffices to prove that $\widetilde{V}(\Omega)$ is a singleton. We prove this by contradiction. Assume that $\widetilde{V}(\Omega)$ is not a singleton. In this case, $\widetilde{V}(\Omega)$ is a polytope with m vertices $(2 \leq m \leq n)$ which we denote by $y_1,\ldots,y_m \in X$. We associate to each vertex y_i a set-valued function $b_i : X^n \rightrightarrows \mathcal{N}$ identifying the agents located at that vertex

$$b_i(x) = \{j \in \mathcal{N} : x_j = y_i\} \qquad \forall\, x \in X^n \qquad \forall\, i = 1,\ldots,m. \tag{25}$$

Observation (24) implies that

$$b_i(x) \neq \emptyset \qquad \forall\, x \in \Omega \qquad \forall\, i = 1,\ldots,m. \tag{26}$$

We arrive at a contradiction with the aid of the following result which holds if the connectivity condition of Theorem 3 is satisfied and $\widetilde{V}(\Omega)$ is not a singleton.

Claim 1: For every $x \in \Omega$ with $b_i(x) \neq \emptyset$ for all $i \in \{1,\ldots,m\}$, there exists $\overline{x} \in \Omega$ with $b_i(\overline{x}) \subseteq b_i(x)$ for all $i \in \{1,\ldots,m\}$ and $b_i(\overline{x}) \subset b_i(x)$ for some $i \in \{1,\ldots,m\}$.

Indeed, a repetitive application of this result eventually leads to the existence of $x^* \in \Omega$ with $b_i(x^*) = \emptyset$ for some $i \in \{1,\ldots,m\}$, contradicting (26). We have thus shown, by contradiction, that $\widetilde{V}(\Omega)$ is a singleton and thus that ζ converges to one of the equilibrium solutions $x_1(t) \equiv \cdots \equiv x_n(t) \equiv$ constant.

We end the proof of Theorem 3 with a proof of Claim 1. Consider an arbitrary $x \in \Omega$ with $b_i(x) \neq \emptyset$ for all $i \in \{1,\ldots,m\}$. There is, by definition, a sequence of times t_q $(q \in \mathbb{N})$ tending to infinity such that $\zeta(t_q) \to x$ as $q \to \infty$. Based upon the sequence of times t_q we construct another sequence of times $\hat{t}_q$ $(q \in \mathbb{N})$ tending to infinity, as follows. Let for each $q \in \mathbb{N}$ the time $\hat{t}_q$ be defined as the first time greater than or equal to t_q at which communication occurs between an agent $k \in b_i(x)$ and an agent $l \notin b_i(x)$ for some $i \in \{1,\ldots,m\}$

$$\hat{t}_q = \min\{t \in [t_q,\infty) : \mathrm{Neighbors}(b_i(x),\ \mathcal{A}(t)) \neq \emptyset \text{ for some } i = 1,\ldots,m\} \qquad \forall\, q \in \mathbb{N}.$$

The validity of this construction follows from the connectivity condition of Theorem 3 that we are assuming to hold. We may

[6]The observation that compactness may be invoked is not trivial, since the set of equilibrium points is unbounded, but follows from the boundedness property of (3) established in Theorem 1.

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assume without loss of generality (by considering an appropriate subsequence if necessary) that

- the $\zeta(\hat{t}_q)$ converge to a limit point $\hat{x} \in \Omega$ as $q \to \infty$ (by boundedness of the solution ζ);
- the $\mathcal{A}(\hat{t}_q)$ are all equal, say, $\mathcal{A}(\hat{t}_q) = \widehat{\mathcal{A}}$ for all $q \in \mathbb{N}$ (since there is only a finite number of possible communication graphs);
- the $\zeta(\hat{t}_q+1)$ converge to a limit point $\overline{x} \in \Omega$ as $q \to \infty$ (by continuity of the set-valued map $e(\widehat{\mathcal{A}})$ and compactness of $e(\widehat{\mathcal{A}})(\hat{x})$ implied by Assumption 1).

It follows that $\overline{x} \in e(\widehat{\mathcal{A}})(\hat{x})$. By construction of the sequence of times $\hat{t}_q$, we conclude that, on the one hand, $b_i(\overline{x}) \subseteq b_i(\hat{x}) = b_i(x)$ for all $i \in \{1, \ldots, m\}$ and, on the other hand, $b_i(\overline{x}) \subset b_i(\hat{x}) = b_i(x)$ for some $i \in \{1, \ldots, m\}$ (by Assumption 1). This concludes the proof of claim 1 and hence also of Theorem 3. □

VI. Conclusion

We have considered in the paper a simple but appealing model for n agents interacting via time-dependent communication links. The model finds wide application in a variety of fields including synchronization, swarming, and distributed decision making. In the model, each agent updates his state based upon the information received from other agents, according to a simple rule. By considering simple dynamics for the individual agents we are able to focus on the role of the communication topology without having to deal with the additional complications arising from complex dynamical agents.

The analysis starts with the assumption that each agent updates his state according to a strict convex combination of its neighbors' states, an assumption which is satisfied in various examples studied in the literature. This assumption leads to the development of a set-valued Lyapunov theory, with the convex hull of the individual agents' states playing the role of a nonincreasing Lyapunov function. Contrary to what might be expected, the dynamics of the multiagent system turns out to be quite subtle and a blend of graph-theoretic and system theoretic tools is used in order to establish necessary and/or sufficient conditions for convergence.

The strongest result is obtained for the case of bidirectional communication. In that case, it is shown that convergence of the individual agents' states to a common value is guaranteed if, during each time interval of the form $[t_0, \infty)$, each agent sends information to each other agent, either through direct communication or indirectly via intermediate agents.

The case of unidirectional communication is more subtle. It is shown by means of a counterexample that, contrary to what might be expected, convergence of the individual agents' states to a common value is not necessarily guaranteed, even if during each interval of the form $[t_0, \infty)$ there is an agent who sends information to all other agents, either through direct communication or indirectly via intermediate agents. Convergence is proven, however, if a uniform bound is imposed on the time it takes for the information to spread over the network.

The counterexample that is used to show that the agents' states may fail to converge to a common value, even in the presence of communication, points toward an interesting phenomenon. Namely that more information exchange does not necessarily lead to improved convergence and may eventually even lead to a loss of convergence, even for the simple models studied in the present paper. The study of the quantitative relationship between communication topology and speed of convergence remains an interesting area of research; see [27], [28], and [30] for some recent results in that direction.

Appendix I
Stability Definitions

In order to enable a clear and precise formulation of the stability and convergence properties of the discrete-time system (3) we extend the familiar stability concepts of Lyapunov theory to the present framework. Notice that we are interested in the agents' states converging to a common, constant value and that we expect this common value to depend continuously on the initial states. In other words, we are dealing with a *continuum* of equilibrium points. This means that the classical stability concepts developed for the study of *individual*, typically isolated, equilibria are not well-adapted to the present situation. Alternatively, one may shift attention away from the individual equilibria and consider the stability properties of the *set* of equilibria instead. The stability of invariant sets has been discussed, for example, in [42] and [43]; see also [44]. Even though set stability indeed provides a valuable framework, we do not proceed along those lines because set stability is not able to fully capture the convergence properties that we are aiming at. For example, even though set stability may be used to assert that the individual agents' states converge to a common value, this common value is not guaranteed to be constant in time and may eventually even grow unbounded; compare with Example 5. This is related to the well-known phenomenon that a trajectory may converge to a set of equilibria without converging to any of the individual equilibria. The stability notions that we introduce below incorporate, by definition, the requirement that all trajectories converge to one of the equilibria.

In the following definition, we make a conceptual distinction between equilibrium solutions and equilibrium points: an equilibrium point is an element of the state space which is the constant value of an equilibrium solution. By referring explicitly to equilibrium *solutions* in the following definition we distinguish the present stability concepts from the more familiar set stability concepts [42]–[44].

Definition 5 (Stability): Let $\mathcal{X}$ be a finite-dimensional Euclidean space and consider a continuous map $f : \mathbb{N} \times \mathcal{X} \to \mathcal{X}$ giving rise to the discrete-time system

$$x(t+1) = f(t, x(t)). \tag{27}$$

Consider a collection of equilibrium solutions of (27) and denote the corresponding set of equilibrium points by Φ. With respect to the considered collection of equilibrium solutions, the dynamical system (27) is called

1) stable if for each $\phi_1 \in \Phi$, for all $c_2 > 0$ and for all $t_0 \in \mathbb{N}$ there is $c_1 > 0$ such that every solution ζ of (27) satisfies: if $|\zeta(t_0) - \phi_1| < c_1$ then there is $\phi_2 \in \Phi$ such that $|\zeta(t) - \phi_2| < c_2$ for all $t \geq t_0$;
2) bounded if for each $\phi_1 \in \Phi$, for all $c_1 > 0$ and for all $t_0 \in \mathbb{N}$ there is $c_2 > 0$ such that every solution ζ of (27)

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satisfies: if $|\zeta(t_0) - \phi_1| < c_1$ then there is $\phi_2 \in \Phi$ such that $|\zeta(t) - \phi_2| < c_2$ for all $t \geq t_0$;

3) globally attractive if for each $\phi_1 \in \Phi$, for all $c_1, c_2 > 0$ and for all $t_0 \in \mathbb{N}$ there is $T \geq 0$ such that every solution ζ of (27) satisfies: if $|\zeta(t_0) - \phi_1| < c_1$ then there is $\phi_2 \in \Phi$ such that $|\zeta(t) - \phi_2| < c_2$ for all $t \geq t_0 + T$;
4) globally asymptotically stable if it is stable, bounded and globally attractive.

Remark 4: We may take without loss of generality $\phi_2 = \phi_1$ in items 1) and 2) of Definition 5 [but not in item 3)]. Compare with the proof of Theorem 4. We have chosen not to incorporate $\phi_2 = \phi_1$ in items 1) and 2) in order to make the similarities with item 3) more transparent.

Definition 5 may be interpreted as follows. Stability and boundedness require that any solution of (27) which is initially close to Φ remains close to one of the equilibria in Φ, thus excluding, for example, the possibility of drift along the set Φ. Global attractivity implies that every solution of (27) converges to one of the equilibria in Φ. If the collection of equilibrium solutions is a singleton consisting of one equilibrium solution, then the notions of stability, boundedness, global attractivity and global asymptotic stability of Definition 5 coincide with the classical notions that have been introduced for the study of individual equilibria. In general, however, the notions introduced here are strictly stronger than their respective counterparts from set stability theory, as is illustrated by the following example.

Example 5: Consider the discrete-time system on $\mathbb{R}^2$

$$x_1(t+1) = \frac{x_1(t)}{\sqrt{1 + (x_1(t))^2}} \tag{28}$$

$$x_2(t+1) = x_1(t) + x_2(t) \tag{29}$$

with $t \in \mathbb{N}$. This system has a continuum of equilibrium points $\{(x_1, x_2) \in \mathbb{R}^2 : x_1 = 0\}$ and it is easy to see that this invariant set is globally asymptotically stable. Nevertheless, the solution of this system starting in $(x_1, x_2) = (1, 0)$ at time $t = 1$ is given by $x_1(t) = 1/\sqrt{t}$ and $x_2(t) = \sum_{\sigma=1}^{t-1} 1/\sqrt{\sigma}$ for all $t > 1$. The x_2-component of this solution diverges to infinity as $t \to \infty$. Accordingly, the collection of equilibrium solutions is not globally asymptotically stable in the sense of Definition 5.

Slightly stronger stability notions result when uniformity with respect to initial time is introduced in Definition 5. If the number c_1 (respectively c_2 and T) may be chosen independently of t_0 in item 1) [respectively items 2) and 3)] then the dynamical system (27) is called uniformly stable (respectively, uniformly bounded and uniformly globally attractive) with respect to the considered collection of equilibrium solutions.

Theorem 4 provides a sufficient condition for uniform stability, uniform boundedness and uniform global asymptotic stability in terms of the existence of a set-valued Lyapunov function. This result is convenient since, on the one hand, set-valued Lyapunov functions arise naturally within the present context as illustrated by Lemma 1, and on the other hand, the stability notions that are asserted in Theorem 4 are precisely those that we are aiming to prove. Apart from its application in the present paper, Theorem 4 may be of independent interest.

Theorem 4 (Lyapunov Characterization): Let $\mathcal{X}$ be a finite-dimensional Euclidean space and consider a continuous map $f : \mathbb{N} \times \mathcal{X} \to \mathcal{X}$ giving rise to the discrete-time system (27). Let Ξ be a collection of equilibrium solutions of (27) and denote the corresponding set of equilibrium points by Φ. Consider an upper semicontinuous set-valued function $V : \mathcal{X} \rightrightarrows \mathcal{X}$ satisfying

1) $x \in V(x), \forall\, x \in \mathcal{X}$;
2) $V(f(t, x)) \subseteq V(x), \forall\, t \in \mathbb{N}, \forall\, x \in \mathcal{X}$.

If $V(\phi) = \{\phi\}$ for all $\phi \in \Phi$, then the dynamical system (27) is uniformly stable with respect to Ξ. If $V(x)$ is bounded for all $x \in \mathcal{X}$, then the dynamical system (27) is uniformly bounded with respect to Ξ.

Consider in addition a function $\mu : \mathrm{Image}(V) \to \mathbb{R}_{\geq 0}$ and a lower semicontinuous function $\beta : \mathcal{X} \to \mathbb{R}_{\geq 0}$ satisfying

3) $\mu \circ V : \mathcal{X} \to \mathbb{R}_{\geq 0} : x \mapsto \mu(V(x))$ is bounded on bounded subsets of $\mathcal{X}$;
4) β is positive definite with respect to Φ; that is, $\beta(\phi) = 0$ for all $\phi \in \Phi$ and $\beta(x) > 0$ for all $x \in \mathcal{X} \setminus \Phi$;
5) $5.\ \mu(V(f(t, x))) - \mu(V(x)) \leq -\beta(x), \forall\, t \in \mathbb{N}, \forall\, x \in \mathcal{X}$.

If $V(\phi) = \{\phi\}$ for all $\phi \in \Phi$ and $V(x)$ is bounded for all $x \in \mathcal{X}$ then the dynamical system (27) is uniformly globally asymptotically stable with respect to Ξ.

We briefly comment on the role of the functions V, μ and β in Theorem 4. The set-valued function V plays the role of a Lyapunov function which is nonincreasing (decreasing) along the solutions of (27). The set-valued nature of V is important: unlike a real-valued function, a set-valued function allows for a continuum of minima which are *not comparable* with each other. For this reason, a set-valued Lyapunov function, unlike a real Lyapunov function, may be used to conclude that each trajectory converges to one equilibrium out of a continuum of equilibria. The function μ serves as a measure for the size of the values of V. In this paper, we let $\mu(V(x))$ be the diameter of the set $V(x)$. The function β characterizes the decrease of V along the solutions of (27) as measured in terms of μ. Of course, the proof of Theorem 4 bears similarities with the classical Lyapunov proofs for real-valued Lyapunov functions.

Proof: (Uniform stability) Consider arbitrary $\phi_1 \in \Phi$ and $c_2 > 0$. If $V(\phi_1) = \{\phi_1\}$, then, by upper semicontinuity of V, there is $c_1 > 0$ such that $V(x) \subset \mathcal{B}(\phi_1, c_2)$ for all $x \in \mathcal{B}(\phi_1, c_1)$. Consider arbitrary $t_0 \in \mathbb{N}$ and $x_0 \in \mathcal{B}(\phi_1, c_1)$ and let ζ denote the solution of (27) with $\zeta(t_0) = x_0$. Conditions 1) and 2) of the theorem imply that

$$\zeta(t) \in V(\zeta(t)) \subseteq V(x_0) \subset \mathcal{B}(\phi_1, c_2) \qquad \forall\, t \geq t_0. \tag{30}$$

(Uniform boundedness) Consider arbitrary $\phi_1 \in \Phi$ and $c_1 > 0$. If $V(x)$ is bounded for all $x \in \mathcal{X}$ then, by upper semicontinuity of V, there is $c_2 > 0$ such that $V(x) \subset \mathcal{B}(\phi_1, c_2)$ for all $x \in \mathcal{B}(\phi_1, c_1)$. Consider arbitrary $t_0 \in \mathbb{N}$ and $x_0 \in \mathcal{B}(\phi_1, c_1)$ and let ζ denote the solution of (27) with $\zeta(t_0) = x_0$. Conditions 1) and 2) of the theorem imply that

$$\zeta(t) \in V(\zeta(t)) \subseteq V(x_0) \subset \mathcal{B}(\phi_1, c_2) \qquad \forall\, t \geq t_0. \tag{31}$$

(Uniform global asymptotic stability) We have already proved that, if $V(\phi) = \{\phi\}$ for all $\phi \in \Phi$ and $V(x)$ is bounded for all $x \in \mathcal{X}$, then the dynamical system (27) is uniformly stable and uniformly bounded with respect to Ξ. It remains to prove uniform global attractivity with respect to Ξ.

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Consider arbitrary $\phi_1 \in \Phi$ and $c_1 > 0$. If $V(x)$ is bounded for all $x \in \mathcal{X}$ then, by upper semicontinuity of V, there is a compact set $K \subset \mathcal{X}$ such that $V(x) \subseteq K$ for all $x \in \mathcal{B}(\phi_1, c_1)$. Similarly as before, Conditions 1) and 2) of the theorem imply that every solution of (27) initiated in $\mathcal{B}(\phi_1, c_1)$ remains in K.

Consider in addition arbitrary $c_2 > 0$. If $V(\phi) = \{\phi\}$ for all $\phi \in \Phi$ then, by upper semicontinuity of V, there is $c_3 > 0$ such that for all $x \in \mathcal{B}(\Phi \cap K, c_3)$ there is $\phi_2 \in \Phi$ such that $V(x) \subset \mathcal{B}(\phi_2, c_2)$. Similarly as before, Conditions 1) and 2) of the theorem imply that every solution of (27) entering $\mathcal{B}(\Phi \cap K, c_3)$ remains in a c_2-ball around some equilibrium point $\phi_2 \in \Phi$.

It remains to prove the existence of $T \geq 0$ such that every solution of (27) starting in $\mathcal{B}(\phi_1, c_1)$ cannot remain longer than T subsequent times in K without entering $\mathcal{B}(\Phi \cap K, c_3)$. In agreement with Conditions 3) and 4) of the theorem and the lower semicontinuity of β, we introduce two real numbers

$$M = \sup_{x \in \mathcal{B}(\phi_1, c_1)} \mu(V(x)) < \infty$$

and

$$\Delta = \min_{x \in K \setminus \mathcal{B}(\Phi, c_3)} \beta(x) > 0.$$

Let $T \geq 0$ be such that $T\Delta > M$. Consider arbitrary $t_0 \in \mathbb{N}$ and $x_0 \in \mathcal{B}(\phi_1, c_1)$ and let ζ denote the solution of (27) with $\zeta(t_0) = x_0$. Then, Condition 5) of the theorem implies that for some $t_1 \in [t_0, t_0 + T]$

$$\zeta(t_1) \in \mathcal{B}(\Phi \cap K, c_3)$$

since otherwise $\mu(V(\zeta(t_0 + T)))$ would be smaller than zero, contradicting that μ takes only nonnegative values. Putting everything together, we conclude that for some $\phi_2 \in \Phi$

$$\zeta(t) \in V(\zeta(t)) \subseteq V(\zeta(t_1)) \subset \mathcal{B}(\phi_2, c_2) \qquad \forall\, t \geq t_0 + T. \tag{32}$$

□

Remark 5: The strict decrease condition (Condition 5) of Theorem 4 may be considerably relaxed. Consider, for example, the following condition which requires that $\mu \circ V$ decreases over time-intervals of length τ.

6) There is a time $\tau \in \mathbb{N}$ such that

$$\begin{aligned}\mu(V(f(t+\tau-1, \ldots f(t+1, f(t,x)) \ldots))) - \mu(V(x)) \\ \leq -\beta(x) \qquad \forall\, t \in \mathbb{N} \qquad \forall\, x \in \mathcal{X}.\end{aligned} \tag{33}$$

Theorem 4 is still true if Condition 5) is replaced by Condition 6).

Remark 6: The Lyapunov functions featuring in Theorem 4 are set-valued functions $V : \mathcal{X} \rightrightarrows \mathcal{X}$, or equivalently, single-valued functions $V : \mathcal{X} \to \mathcal{P}(\mathcal{X})$. One may be interested in generalizations of Theorem 4 considering more abstract Lyapunov functions $V : \mathcal{X} \to Z$ with Z not necessarily equal to $\mathcal{P}(\mathcal{X})$. The determination of appropriate structures and properties for the set Z and the function V enabling Lyapunov-type of results is beyond the scope of this study. Nevertheless, we point out that a partial ordering of the set Z seems to be an essential ingredient in order to be able to conclude that every solution converges to one equilibrium out of a continuum of equilibria. It turns out that the class of set-valued functions $V : \mathcal{X} \rightrightarrows \mathcal{X}$ is universal in this context, in the following sense. Consider a set Z and let $\prec$ by a strict partial ordering of Z.[7] Consider a set $\mathcal{X}$ and a function $\widetilde{V} : \mathcal{X} \to Z$. Introduce for every $x \in \mathcal{X}$ the subset $V(x) \subseteq \mathcal{X}$ defined by

$$V(x) = \{\hat{x} \in \mathcal{X} : \widetilde{V}(\hat{x}) \prec \widetilde{V}(x) \text{ or } \widetilde{V}(\hat{x}) = \widetilde{V}(x)\}.$$

Then, we have $V(x) = V(\overline{x})$ whenever $\widetilde{V}(x) = \widetilde{V}(\overline{x})$ and $V(x) \subset V(\overline{x})$ whenever $\widetilde{V}(x) \prec \widetilde{V}(\overline{x})$. The proof of this statement is elementary and, therefore, omitted. Not only does this result provide some motivation for restricting attention to set-valued maps $V : \mathcal{X} \rightrightarrows \mathcal{X}$ in Theorem 4, it also points toward a constructive method for finding such set-valued Lyapunov functions; see, for example, the proofs of Theorems 1 and 2.

Appendix II
Graph-Theoretic Result

Theorem 5: A directed graph $(\mathcal{N}, \mathcal{A})$ has a node connected to all other nodes if and only if every pair of nonempty, disjoint subsets $\mathcal{L}_1, \mathcal{L}_2 \subset \mathcal{N}$ satisfies $\text{Neighbors}(\mathcal{L}_1, \mathcal{A}) \cup \text{Neighbors}(\mathcal{L}_2, \mathcal{A}) \neq \emptyset$.

Independently of the current research, this result is also proven in [22, Lemma 1].

Proof: (Only if) Assume that the node k is connected to all other nodes, and consider an arbitrary pair of nonempty, disjoint subsets $\mathcal{L}_1, \mathcal{L}_2 \subset \mathcal{N}$. We distinguish three cases.

1) If $k \in \mathcal{L}_1$, then $\text{Neighbors}(\mathcal{L}_2, \mathcal{A})$ is nonempty.
2) If $k \in \mathcal{L}_2$, then $\text{Neighbors}(\mathcal{L}_1, \mathcal{A})$ is nonempty.
3) If $k \notin \mathcal{L}_1 \cup \mathcal{L}_2$ then $\text{Neighbors}(\mathcal{L}_1, \mathcal{A})$ and $\text{Neighbors}(\mathcal{L}_2, \mathcal{A})$ are both nonempty.

In all three cases $\text{Neighbors}(\mathcal{L}_1, \mathcal{A}) \cup \text{Neighbors}(\mathcal{L}_2, \mathcal{A}) \neq \emptyset$.

(If) The proof consists of a constructive algorithm that is guaranteed to terminate in a finite number of steps. Each step $i \in \mathbb{N}$ (except the last step) involves the selection of four nonempty sets $\mathcal{L}_1(i) \subseteq \mathcal{F}_1(i) \subset \mathcal{N}$ and $\mathcal{L}_2(i) \subseteq \mathcal{F}_2(i) \subset \mathcal{N}$ satisfying:

- $\mathcal{F}_1(i)$ and $\mathcal{F}_2(i)$ are disjoint;
- each node in $\mathcal{L}_1(i)$ is connected to each other node in $\mathcal{F}_1(i)$ and each node in $\mathcal{L}_2(i)$ is connected to each other node in $\mathcal{F}_2(i)$.

Step 1): Set $\mathcal{L}_1(1) = \mathcal{F}_1(1) = \{m_1\}$ and $\mathcal{L}_2(1) = \mathcal{F}_2(1) = \{\overline{m}_1\}$, where m_1 and $\overline{m}_1$ are two arbitrary, different nodes of the graph. (Here, we have assumed that the graph has at least 2 nodes. If the graph has only one node, then the statement of Theorem 5 is trivial.)

Step $i > 1$): As we are proving the *if*-part of the theorem, we may assume that $\text{Neighbors}(\mathcal{L}_1(i-1), \mathcal{A}) \cup \text{Neighbors}(\mathcal{L}_2(i-1), \mathcal{A}) \neq \emptyset$. We restrict attention to the case $\text{Neighbors}(\mathcal{L}_2(i-1), \mathcal{A}) \neq \emptyset$ and consider $m_i \in \text{Neighbors}(\mathcal{L}_2(i-1), \mathcal{A})$. (The alternative case of $\text{Neighbors}(\mathcal{L}_1(i-1), \mathcal{A}) \neq \emptyset$ can be dealt with in a completely similar way, be interchanging the role of $\mathcal{L}_1$ and $\mathcal{L}_2$, respectively $\mathcal{F}_1$ and $\mathcal{F}_2$.) We distinguish four cases.

1) If $m_i \in \mathcal{F}_1(i-1)$ and $\mathcal{F}_1(i-1) \cup \mathcal{F}_2(i-1) = \mathcal{N}$, then the algorithm terminates: any agent $\overline{m}_i \in \mathcal{L}_1(i-1)$ is connected to all other agents in the graph.

[7] A partial ordering $\prec$ of a set Z is a transitive and antisymmetric relation on Z. If we never have $z \prec z$ then $\prec$ is called a strict partial ordering.

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2) If $m_i \in \mathcal{F}_1(i-1)$ and $\mathcal{F}_1(i-1) \cup \mathcal{F}_2(i-1) \neq \mathcal{N}$, then set

$$\mathcal{L}_1(i) = \mathcal{L}_1(i-1) \tag{34}$$
$$\mathcal{F}_1(i) = \mathcal{F}_1(i-1) \cup \mathcal{F}_2(i-1) \tag{35}$$
$$\mathcal{L}_2(i) = \mathcal{F}_2(i) = \{\overline{m}_i\} \tag{36}$$

where $\overline{m}_i$ is an arbitrary node which does not belong to $\mathcal{F}_1(i-1) \cup \mathcal{F}_2(i-1)$.

3) If $m_i \notin \mathcal{F}_1(i-1) \cup \mathcal{F}_2(i-1)$, then set

$$\mathcal{L}_1(i) = \mathcal{L}_1(i-1) \tag{37}$$
$$\mathcal{F}_1(i) = \mathcal{F}_1(i-1) \tag{38}$$
$$\mathcal{L}_2(i) = \{m_i\} \tag{39}$$
$$\mathcal{F}_2(i) = \mathcal{F}_2(i-1) \cup \{m_i\}. \tag{40}$$

4) If $m_i \in \mathcal{F}_2(i-1) \setminus \mathcal{L}_2(i-1)$ then

$$\mathcal{L}_1(i) = \mathcal{L}_1(i-1) \tag{41}$$
$$\mathcal{F}_1(i) = \mathcal{F}_1(i-1) \tag{42}$$
$$\mathcal{L}_2(i) = \mathcal{L}_2(i-1) \cup \{m_i\} \tag{43}$$
$$\mathcal{F}_2(i) = \mathcal{F}_2(i-1). \tag{44}$$

The algorithm is guaranteed to end in a finite number of steps, since at each step (except the last step) either the number of agents in $\mathcal{F}_1 \cup \mathcal{F}_2$ increases, or the number of agents in $\mathcal{F}_1 \cup \mathcal{F}_2$ remains unchanged and the number of agents in $\mathcal{L}_1 \cup \mathcal{L}_2$ increases. □

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